Inorganic Cume Nov. 17,2000

1) Discuss the following acid/base definitions. Include a balanced reaction that illustrates the definition, labeling the acid and base. Point out the chemical or electronic characteristics shared by all acids and bases in the model. Briefly comment on the relationship(s) between the various models.

a) Bronsted-Lowry
b) Arrhenius
c) Lewis
d) Frankland or Solvent System

2) Qualitative or semi-quantitative models for the prediction of acid or base strength have been proposed by several investigators. Discuss the basis for the following models.

a) Drago-Wyland
b) HSAB
c) Pauling

3) Why do gas phase proton affinities for bases differ from aqueous solution base strength measures such as K_b values?

4) Account for the following experimentally observed K_a trends.

a) For the dicarboxylic acid series $HO_2C(CH_2)_nCO_2H$

n	1	2	3	4	5	6	7	8
K_{a1}/K_{a2}	1120	29.5	17.4	12.3	11.2	10.0	9.5	9.3

b)

	H_3AsO_3	H_3BO_3	H_3PO_3
pK_{a1}	9.22	9.22	1.8
pK_{a2}	***	***	6.15
pK_{a3}	***	***	***

*** K_a too small to measure in aqueous solution